Basic Enterprise Network Design Using Switches and Firewall

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Introduction

This project demonstrates the design and implementation of a basic enterprise network topology using switches, end devices, and a firewall. The network was created in a network simulation environment to apply practical networking concepts such as switching, network segmentation, and secure communication.

The project focuses on creating a simple and organized infrastructure that allows communication between devices while maintaining security using a firewall.

Project Objectives

- Build a basic enterprise network topology.
- Connect end devices using switches.
- Secure the network using a firewall.
- Understand basic switching concepts.
- Improve communication between network devices.
- Apply practical networking skills.

Network Components

1 . Firewall

The network includes one firewall device:

Device Name	Model
FW1	USG6000V

Firewall Functions

- Protect the network from unauthorized access.
- Filter incoming and outgoing traffic.
- Control communication between network segments.
- Improve overall network security.

2. Switches

Two switches are used in the topology.

Switch Name	Model
LSW1	S3700
LSW2	S3700

Functions of the Switches

- Connect end devices to the network.
- Forward data between connected devices.
- Improve network organization.
- Support communication within the LAN.

3. End Devices

The project contains two PCs.

Device Name	Type
PC1	PC
PC2	PC

Purpose of End Devices

- Test network connectivity.
- Simulate user communication.
- Access network resources.

Network Topology

The firewall is connected to both switches, and each switch is connected to a separate PC.

Connection Structure

- FW1 connected to LSW1
- FW1 connected to LSW2
- LSW1 connected to PC1
- LSW2 connected to PC2

This structure allows secure communication between devices through the firewall.

Network Operation

1. PCs connect to their respective switches.
2. Switches forward traffic within the network.
3. The firewall controls and secures communication.
4. Devices can communicate according to firewall rules and network configuration.

Security Features

- Firewall- based traffic filtering.
- Controlled communication between devices.
- Network isolation and protection.
- Prevention of unauthorized access.

Advantages of the Project

- Simple and organized design.
- Easy to manage and troubleshoot.
- Provides basic network security.
- Demonstrates practical networking concepts.
- Suitable for beginners learning enterprise networking.

Technologies Used

- Switching

- Firewall Security
- LAN Networking
- Network Topology Design
- Basic Enterprise Networking

Possible Improvements

Future improvements can include:

- Adding VLANs for network segmentation.
- Implementing routing between networks.
- Adding wireless connectivity.
- Expanding the number of devices.
- Applying Access Control Lists (ACLs).

Conclusion

This project successfully demonstrates a simple enterprise network infrastructure using switches, PCs, and a firewall. It provides a practical understanding of network communication, switching, and security concepts. The topology can also serve as a foundation for more advanced networking projects in the future.